

ANDY OUYANG

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EDUCATION

Carnegie Mellon University - Pittsburgh, PA

May 2026

Bachelor of Science in Statistics and Machine Learning, Additional Major in Decision Science (GPA: 3.77)

Coursework: Algorithms and Advanced Data Structures, Advanced Data Analysis (R), Modern Regression (R), Introduction to Machine Learning (Python), Art and Machine Learning (Python), Statistical Graphics & Visualizations (R), Principles of Imperative Computation (C), Statistical Computing (R), Fundamentals of Programming (Python)

Activities: Statistical Computing 36-350 Teaching Assistant, Students Using Data for Social Good (SUDS) Project Member, CMU Data Science Club, Taiwanese Student Association

SKILLS

Programming: Python (pandas, Matplotlib, NumPy, PyTorch, TensorFlow), R (ggplot2, tidyverse), SQL, C

Tools: AWS, GCP, Git, Jupyter, Google Analytics, Looker Studio, Microsoft Excel, Qualtrics

EXPERIENCE

New York Post - New York, NY

June 2025 - August 2025

Data and Analytics Intern

- Designed and implemented a Markov chain multi-touch attribution model to evaluate user behavior and optimize marketing spend
- Conducted attribution analyses to identify high-impact user touchpoints, directly informing subscription growth strategies
- Collaborated with product and marketing teams to turn business objectives into data solutions aligned with strategic goals
- Delivered clear, actionable insights through ad hoc analyses and reporting, supporting decision-making on audience engagement and strategic initiatives, utilizing data sources (Google Analytics, JWPlayer) and analysis tools (Excel, AWS Athena)

Carnegie Mellon University, Department of Social and Decision Sciences - Pittsburgh, PA

March 2024 - Present

Undergraduate Research Assistant

- Design and optimize a Qualtrics survey on job search decision-making, integrating AWS Lambda with OpenAI API, improving real-time user interactions and boosting data collection efficiency by 50%
- Apply experimental design principles in pilot testing, refining survey flow and increasing participant satisfaction by 90%
- Support multiple research initiatives by managing survey distribution to participants and overseeing in-person studies

Oakland Business Improvement District - Pittsburgh, PA

June 2023 - February 2024

Placemaking and Special Events Intern

- Led a comparative analysis of visitor metrics across 5+ major cities to inform an Oakland retail marketing strategy project
- Analyzed district visitation trends using Placer.ai data in Python and Excel to assess event impact and business performance
- Developed data visualizations for marketing reports and meetings to deliver actionable insights on district foot traffic trends
- Supported event outreach efforts and documented key discussions through meeting minutes to facilitate team coordination

NYC Department of Small Business Services - New York, NY

July 2022 - August 2022

Regulatory Reform Policy Intern

- Analyzed and summarized surveys, archival data, reports, and policies to evaluate small business regulation challenges
 - Developed and presented a redevelopment proposal with data-driven compliance enforcement policy recommendations to staff
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PROJECTS

The Belief Spiral Explorer - Python | [Project Link](#)

April 2025

- Scraped and processed 1K+ articles from NewsAPI, NYT, Guardian, Reddit, and Wikipedia and cleaned with spaCy keyword extraction and regex/POS filtering to create a structured dataset for semantic analysis and misinformation modeling
- Built a belief graph using NetworkX with weighted co-occurrence edges and trained Node2Vec embeddings to capture semantic similarity, enabling dynamic exploration of interconnected beliefs and conspiracy clusters
- Experimented with Stable Diffusion and fine-tuned LLaMA-2 to prototype AI-driven conspiracy story and image generation

CMU AI Sustainability Datathon 2024 (Honorable Mention) | [GitHub](#)

November 2024

- Researched and analyzed datasets on the environmental impact of AI and cloud computing, using pandas and Matplotlib
- Proposed a cloud-based sustainability solution for AI operations, reducing projected carbon emissions

Traveler's Aid & Students Using Data for Social Good (SUDS)

September 2024 - May 2025

- Designed and implemented a survey on impact of transportation on food insecurity in Qualtrics for an 8-week study
- Conducted literature review and synthesized findings to refine survey design and identify key variables of interest
- Developed analysis plan with ridgeplots, t-tests, KS tests, and regressions to evaluate transportation funding impacts

Public Opinion on Authoritarian Influence and Democracy in Africa - R | [Project Link](#)

May 2024 - July 2024

- Cleaned and examined Afrobarometer data to develop data visualizations to show public opinion trends on democracy
- Reproduced and validated seven regression models from a study on public opinion toward military rule in Mali, using R for statistical analysis and QGIS to integrate geographic data for spatial insights

Analyzing Innovation Dynamics Within K-pop Music - Python & R | [GitHub](#)

April 2024 - May 2024

- Utilized Spotify API to compile a dataset and computed feature vector similarity to analyze musical evolution across albums
- Developed time series visualizations to track lyrical and audio similarity, identifying musical shifts and relation to popularity
- Applied regression analysis to examine the balance between musical innovation and similarity in shaping listener engagement